Infinity
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SUBJECTIVE WORKSHEET

Life Processes in Animals

Science — Biology · Class 7 · Max Marks: 34 · Time: 1 hour

Section A — Very Short Answer (1 × 5 = 5)

1. Name the projections that increase absorption in the small intestine. [1]

2. Name the organ that produces bile. [1]

3. Name the respiratory organ of a fish. [1]

4. What is the partly digested food that ruminants chew again called? [1]

5. Name the process by which absorbed food is used by the body cells. [1]

Section B — Short Answer (2 × 5 = 10)

1. Differentiate between ingestion and egestion. [2]

2. Why is the small intestine long and provided with villi? [2]

3. How does an Amoeba obtain and digest its food? [3]

4. Why can a cow digest grass while a human being cannot? [2]

5. Name any two respiratory organs found in different animals and state one animal that uses each. [2]

Section C — Long Answer (5 × 3 = 15)

1. Describe the five steps of nutrition in animals, writing one line about each. [5]

2. Trace the path of food through the human alimentary canal, naming the main organs and the job of each. [5]

3. Explain how the exchange of gases takes place in humans during breathing. [5]

Section D — Higher-Order Thinking (Give Reasons) (2 × 2 = 4)

1. A person has had part of the small intestine removed. Explain why they may become weak even though they eat enough food. [2]

2. Earthworms breathe through their moist skin. Explain why they come out of the soil during heavy rain. [2]

Answer Key

Very Short Answer

1. Villi.
2. The liver.
3. Gills.
4. Cud.
5. Assimilation.

Short Answer

1. Ingestion is the taking in of food into the body; egestion is the removal of undigested food from the body.
2. Its length and the villi together give a very large surface area, so that the maximum digested food can be absorbed into the blood.
3. It puts out pseudopodia that surround the food and trap it in a food vacuole; the food is digested there and the useful part absorbed into the cell.
4. A cow's stomach holds bacteria that digest the cellulose of grass; humans do not have these bacteria, so they cannot digest cellulose.
5. Lungs — humans/mammals; gills — fish; moist skin — earthworm; tracheae — insects (any two).

Long Answer

1. Ingestion — taking in food; digestion — breaking food into simple, soluble substances; absorption — soluble food passes into the blood; assimilation — cells use the food for energy and growth; egestion — undigested food is removed.
2. Mouth (chewing + saliva) → oesophagus (pushes food) → stomach (churns, acid + juices) → small intestine (digestion completed, food absorbed through villi) → large intestine (water absorbed) → anus (egestion). Liver and pancreas add digestive juices.
3. Air is breathed in through the nose into the lungs and reaches the alveoli. Oxygen from the air passes into the blood, and carbon dioxide passes from the blood into the alveoli and is breathed out.

Higher-Order Thinking (Give Reasons)

1. A shorter small intestine has less surface for absorption, so less digested food enters the blood and the body is undernourished.
2. Rain fills the soil spaces with water, leaving little air, so the earthworms come to the surface to get oxygen.